BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 265 · Operating systems and assembly challenge

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which OS component queues print requests?

Q2. What does LMC INP do to ACC?

Q3. What does Waiting mean?

Q4. Rising-edge D flip-flop has Q=1 and D=0 immediately before the active edge. What is Q after?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

ACC — Accumulator

Ready — Eligible for CPU

Running — Using CPU

Waiting — Blocked on event

New learning

Understand — LMC INP overwrites the accumulator; STA saves the value to a mailbox.

Trace — Processes change between Ready, Running and Waiting as the OS schedules and responds to I/O.

Explain — Spooling enables multiple programs to submit print jobs without monopolising the printer.

Worked example

Trace an LMC fragment with input 8.

1. Predict first — INP sets ACC=8; STA X writes 8 to mailbox X; the ACC remains 8. HLT ends the program.

We Do · Practise with guidance

A Running process requests a disk read. Give its next states.

```
Running → Waiting → Ready → Running
```

Predict first

The process waits for disk completion, then becomes Ready for selection by the scheduler.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

State ACC and X after input 8 followed by INP; STA X. [4]

Question 2

Describe Ready, Running, Waiting and return to Ready during a disk read. [6]

Question 3

Describe OS file access, I/O process state and spooling when an app prints saved data. [6]

Exit Ticket · Answer before checking

A blocked process continues using the CPU until its I/O completes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A print spooler.

Q2. Replaces ACC with supplied input.

Q3. Blocked until an event such as I/O completes.

Q4. Q=0; it samples D at the clock edge.

You Do pages · Independent questions

Q1. ACC=8 after INP [2]; X becomes 8 after STA [1]; ACC remains 8 [1].

Q2. Ready waits CPU [1], Running executes [1], I/O request [1] causes Waiting [1], I/O completion [1] returns Ready [1].

Q3. OS file lookup [1], permission [1], driver [1], possible Waiting [1], spool queue [1], scheduled printer completion [1].

Exit Ticket · Model answer

False. It normally enters Waiting.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: